American International University- Bangladesh

Faculty of Engineering (FE)

Department of Electrical and Electronic Engineering (EEE)

EEE3101: DIGITAL LOGIC AND CIRCUITS LAB

Section: W

Group: 01

Lab Report -03

Design of adder and comparator circuits.

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Title: Design of adder and comparator circuits.

Abstract:

In this experiment, adder and comparator circuits were designed and implemented using basic logic gates. Both half-adder and full-adder circuits were constructed to perform binary addition, and their truth tables were verified. A 2-bit comparator was then developed using cascaded 1-bit comparator blocks to determine the relative magnitude of two binary numbers. The functionality of all circuits was tested through both simulation and hardware implementation. NI Multisim was used for simulation and the observed results were analyzed and compared to ensure accuracy.

Theory:

Adder: Adders and subtractors are digital circuits which are capable of adding and subtracting binary digits. They are the most important part in the design of Arithmetic Logic Unit (ALU). They are of two types –

- **Half Adder:** A half adder is a combinational logic circuit that adds two single-bit binary numbers. It produces two outputs: the Sum (S) and the Carry (C). The Sum is obtained using an XOR gate, and the Carry is obtained using an AND gate. It does not handle carry input from a previous stage.

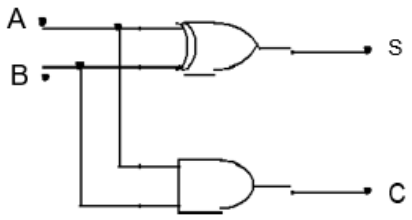


Figure: Half Adder

Boolean expression for Half Adder –

$S = A \oplus B$

$C_{out} = AB$

Truth Table for Half Adder:

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder: A full adder is a combinational circuit that adds three binary inputs: two significant bits and a carry-in from the previous stage. It generates two outputs: Sum (S) and Carry-out (Cout). The Sum is calculated using XOR gates, while the Carry-out is derived using AND and OR gates. Full adders are used to build multi-bit binary adders.

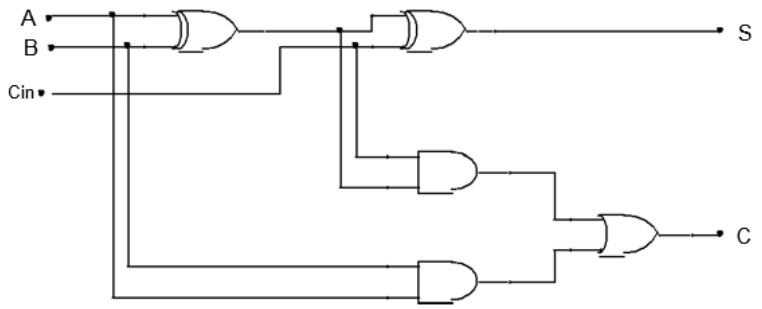


Figure: Full Adder

Boolean expression for Full Adder –

$S = A \oplus B \oplus C_{in}$

$C_{out} = C_{in} (A \oplus B) + AB$

Truth Table for Full Adder:

A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Comparator: A magnitude comparator is a device that takes in two sets of inputs in its input and compares them to provide an output, if they are equal, greater than or less than the other.

- **1-Bit Comparator:** A 1-bit comparator is a combinational circuit that compares two single-bit binary inputs, A and B. It produces three outputs indicating whether A is greater than, less than, or equal to B. Logic gates are used to derive the conditions: A=B (when both inputs are the same), A>B (when A=1 and B=0), and A<B (when A=0 and B=1). These outputs help determine the relative magnitude of the inputs.

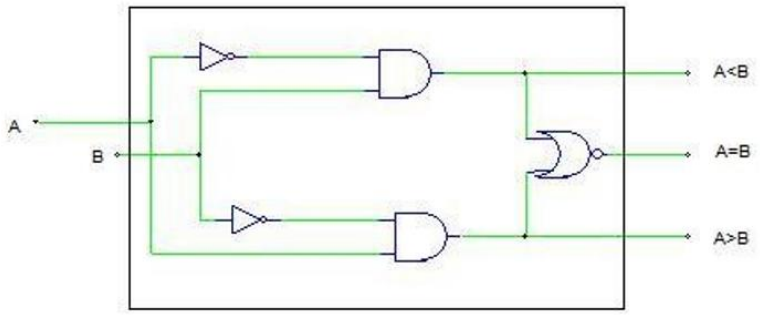


Figure: 1-Bit Comparator

Truth Table for 1-Bit Comparator:

A	B	A = B	A > B	A < B
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

- **2-Bit Comparator:** A 2-bit comparator is a digital circuit designed to compare two 2-bit binary numbers and determine their relative magnitude. It evaluates the higher-order bits first (A1 vs. B1) to decide; if they are equal, it then compares the lower-order bits (A0 vs. B0). Based on this comparison, it generates three outputs indicating whether the first number is greater than, less than, or equal to the second. The circuit is typically built using multiple logic gates or by combining 1-bit comparator modules.

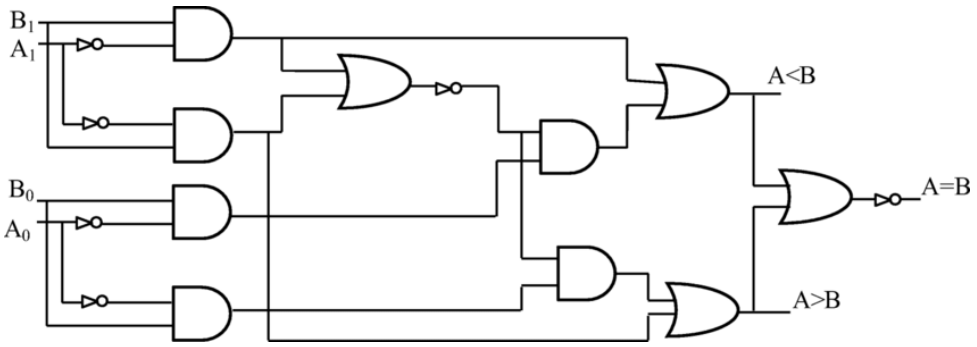


Figure: 2-Bit Comparator

Truth Table for 2-Bit Comparator:

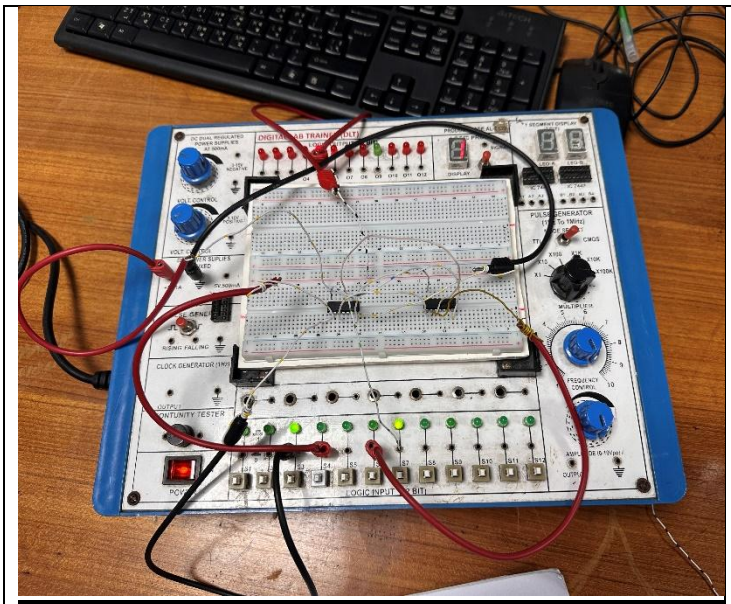
Inputs				**Outputs**		
A1	**A0**	**B1**	**B0**	**A = B**	**A > B**	**A < B**
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

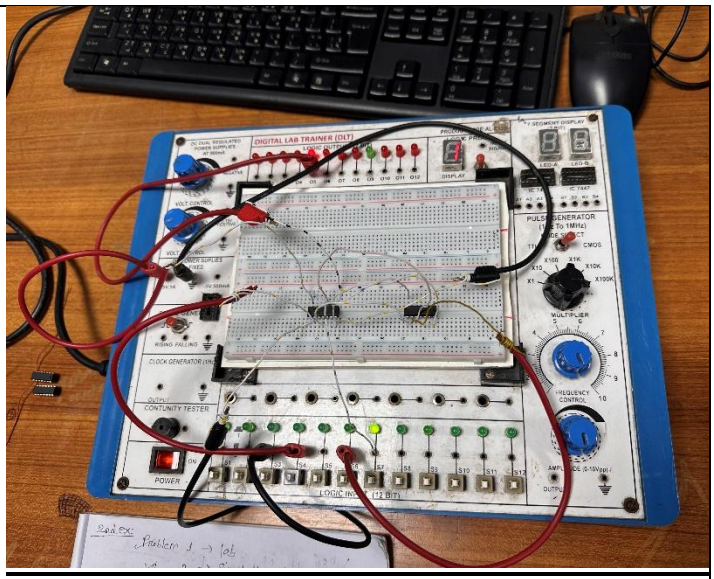
Apparatus:

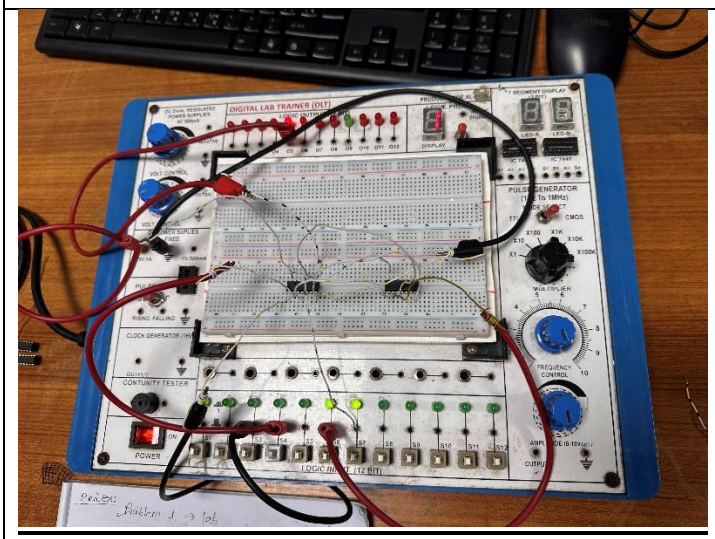
1. Digital trainer board
2. IC 7408:2pcs
3. IC 7404:2pcs
4. IC 7486:2pcs
5. IC 7431:2pcs
6. IC 7483:1pcs
7. Connecting wires

Methodology and Analysis:

Experiment Images:







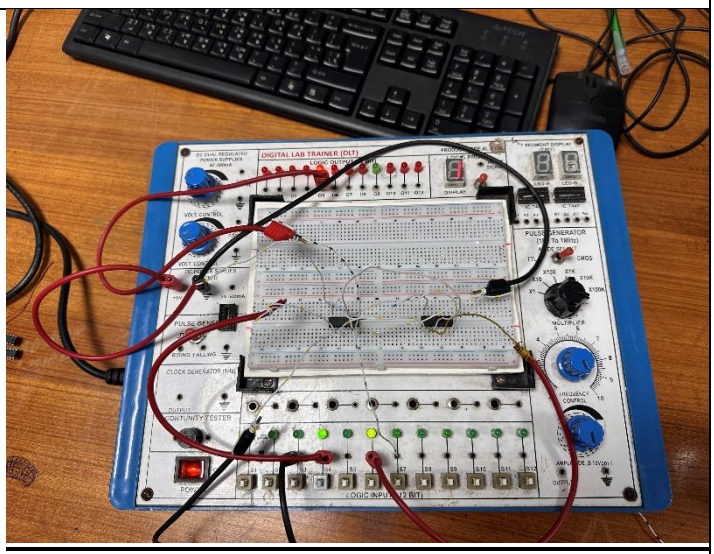


Fig:Full Adder

Simulations:

Half Adder:

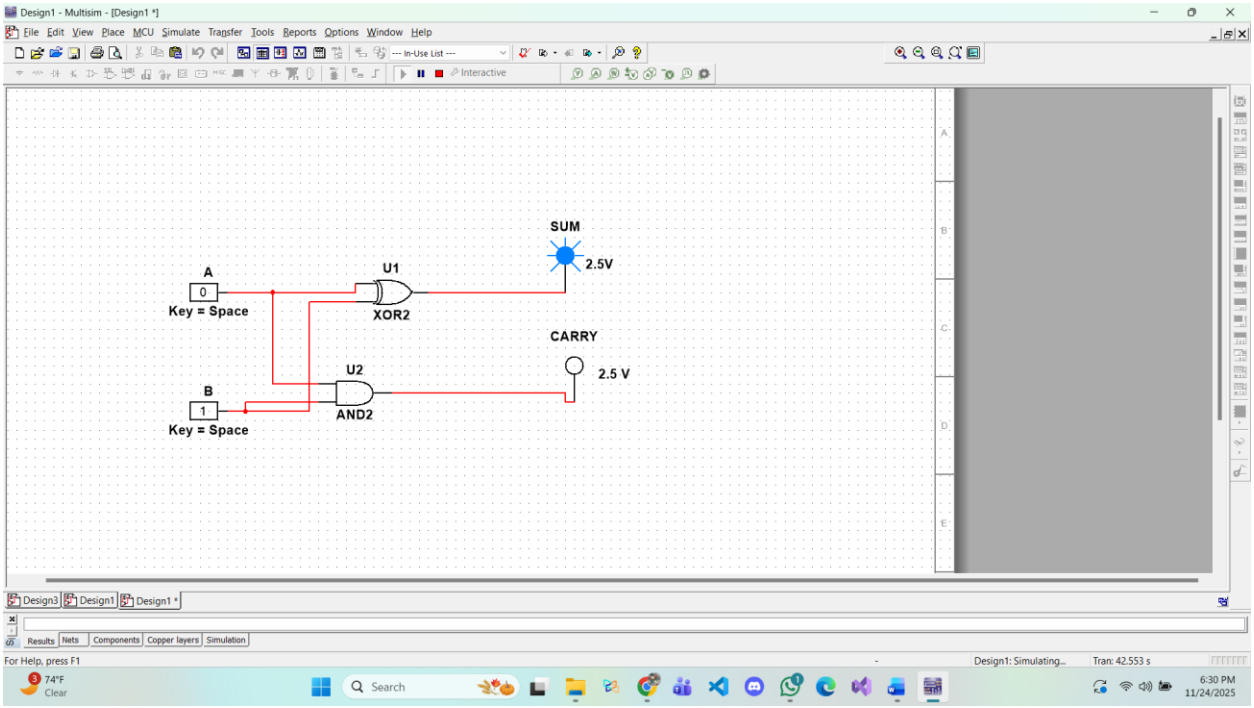


Fig: Half Adder (0+1->S=1;C=0)

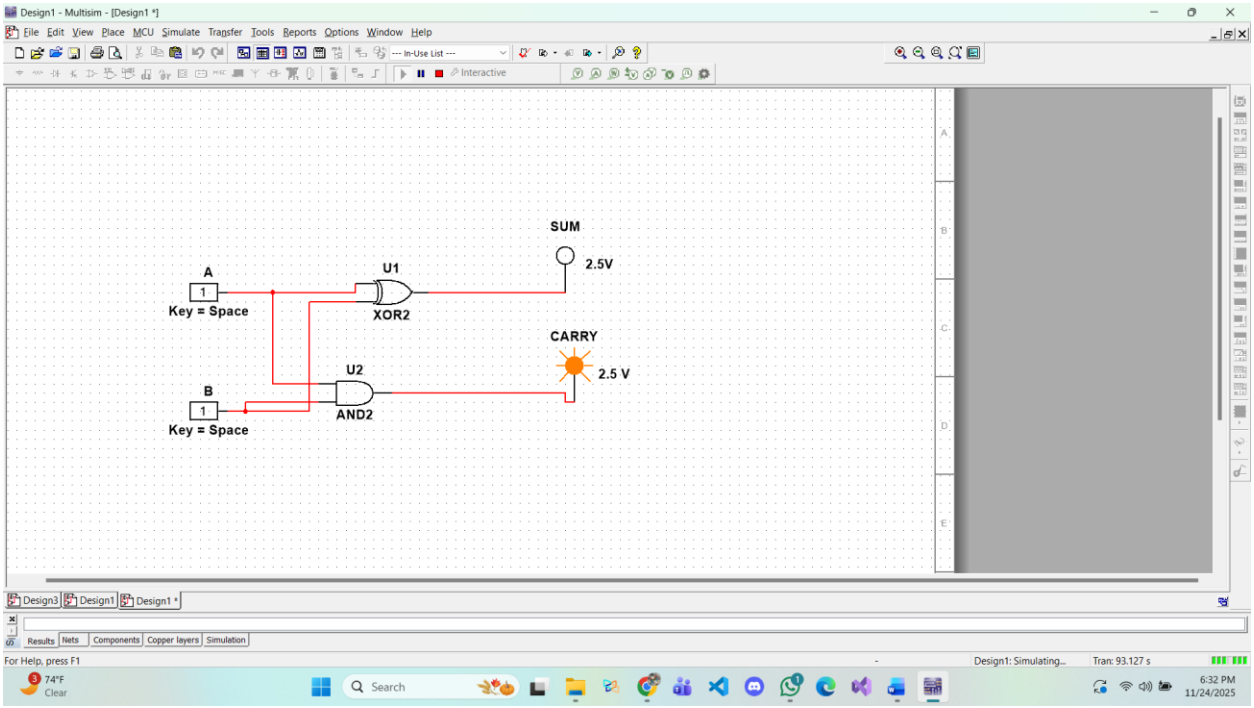


Fig: Half Adder (1+1->S=0;C=1)

Full Adder:

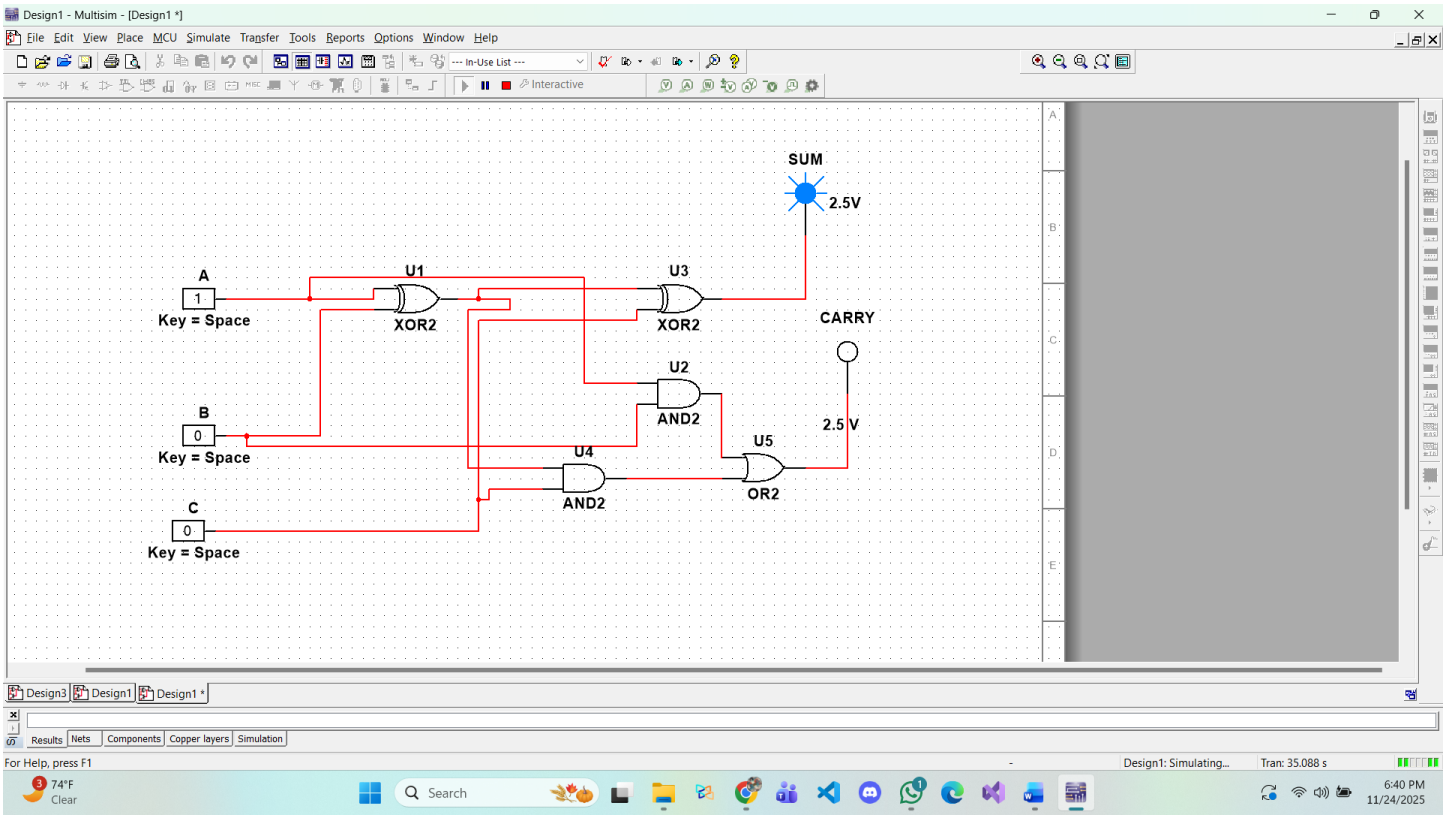


Fig:Full Adder (1+0+0->S=1;C=0)

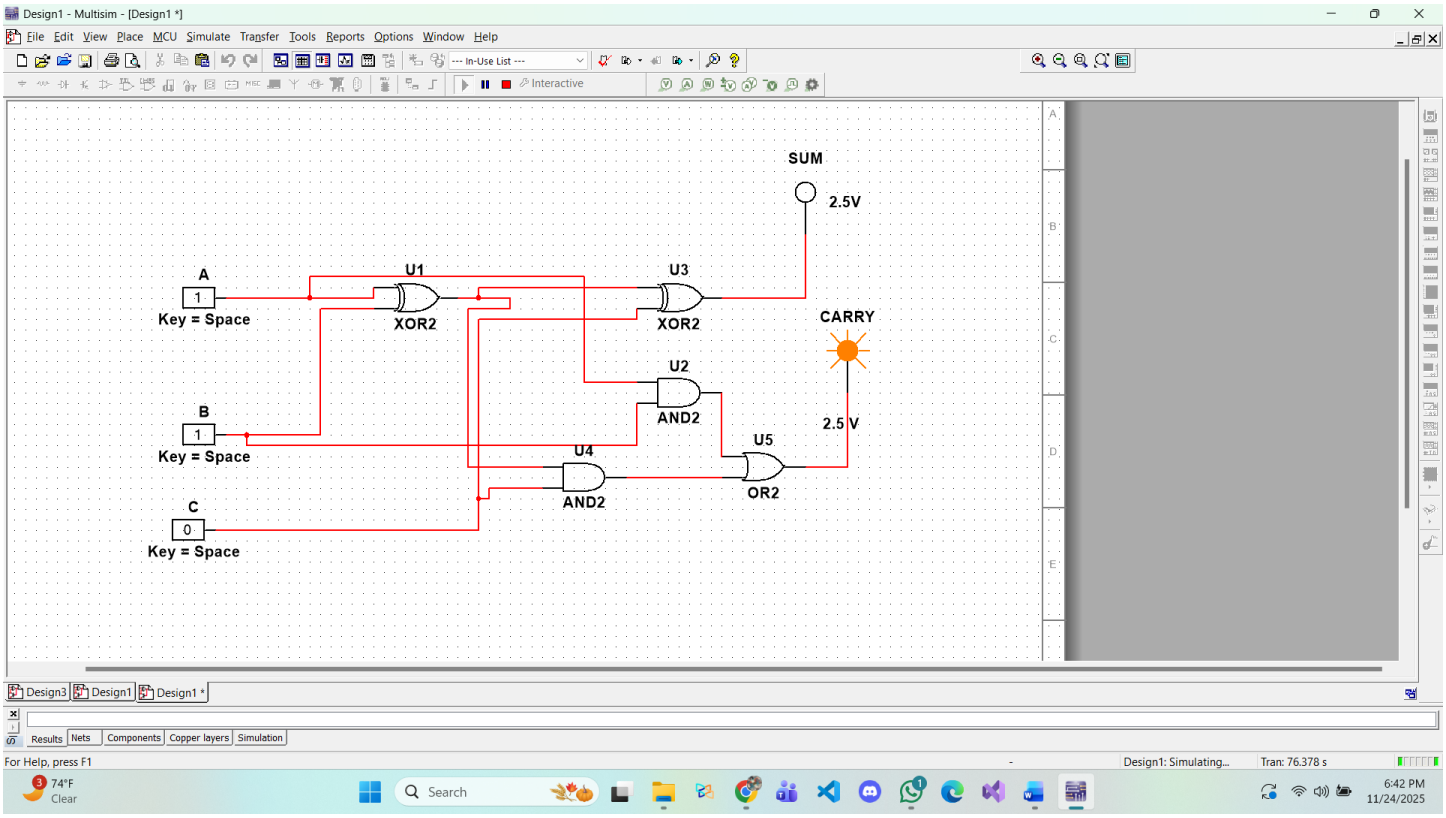


Fig:Full Adder (1+1+0->S=0;C=1)

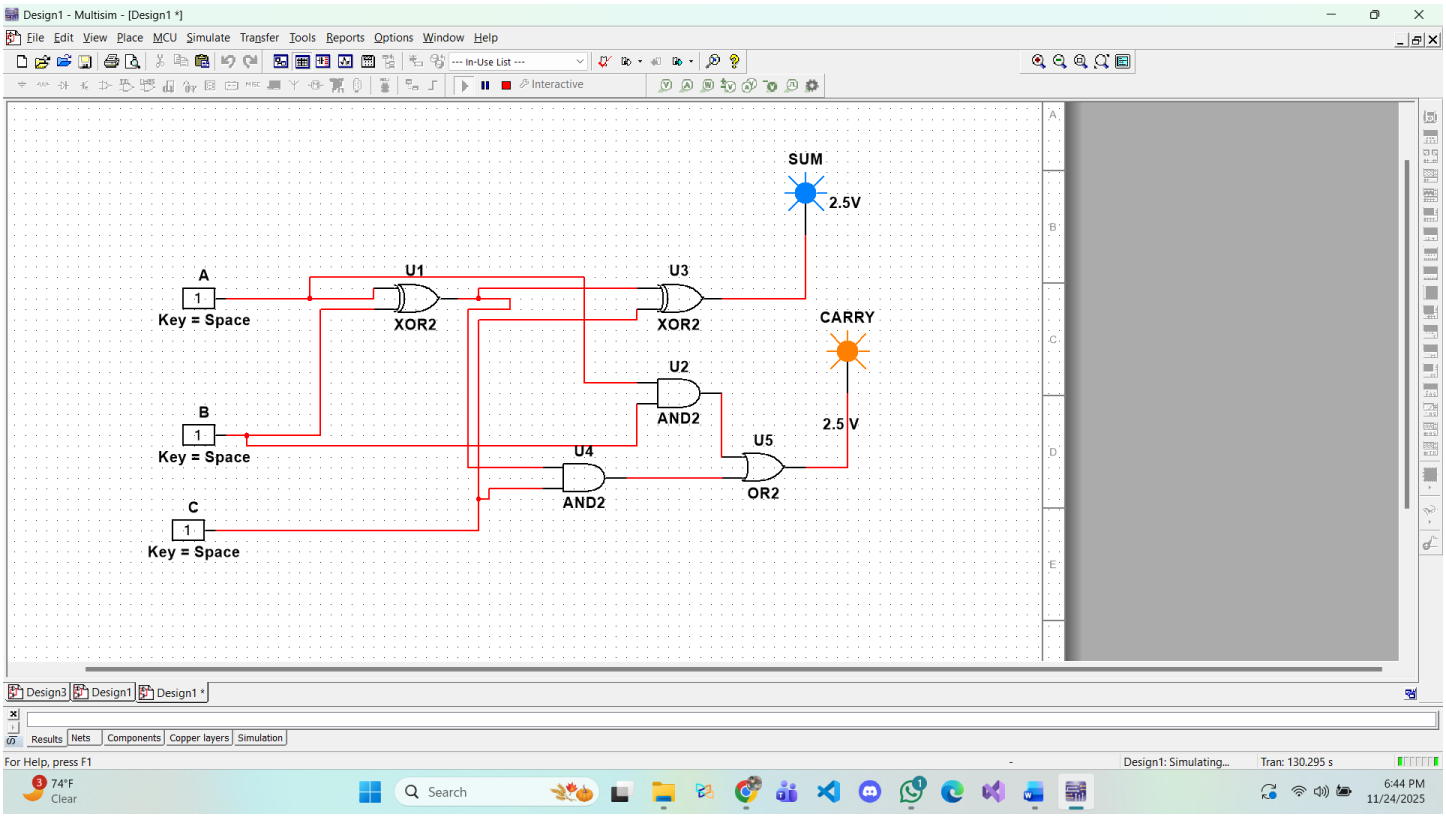


Fig:Full Adder (1+1+1->S=1;C=1)

1 bit Comparator:

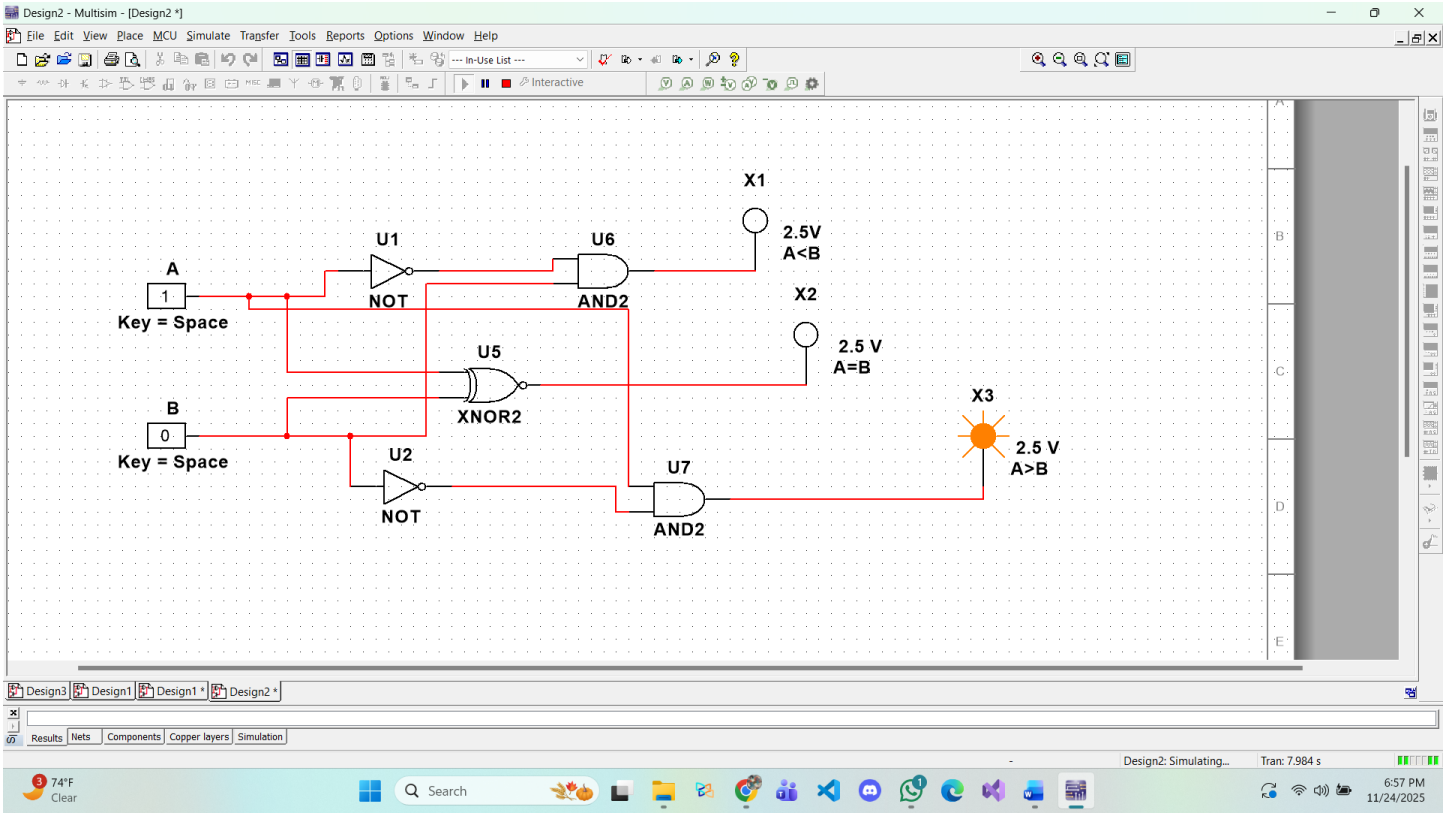


Fig: 1bit Comparator (A>B)

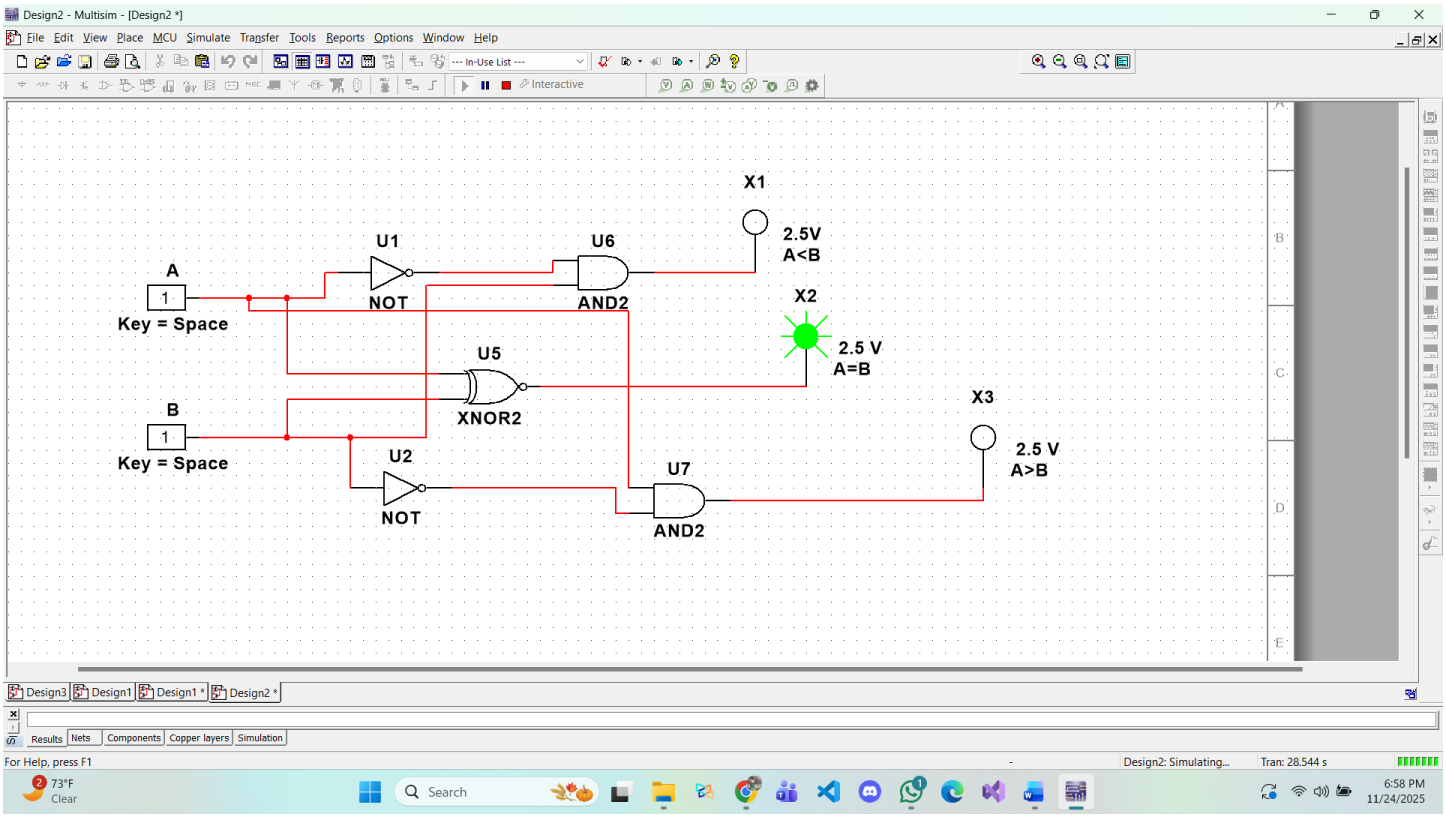


Fig: 1bit Comparator (A=B)

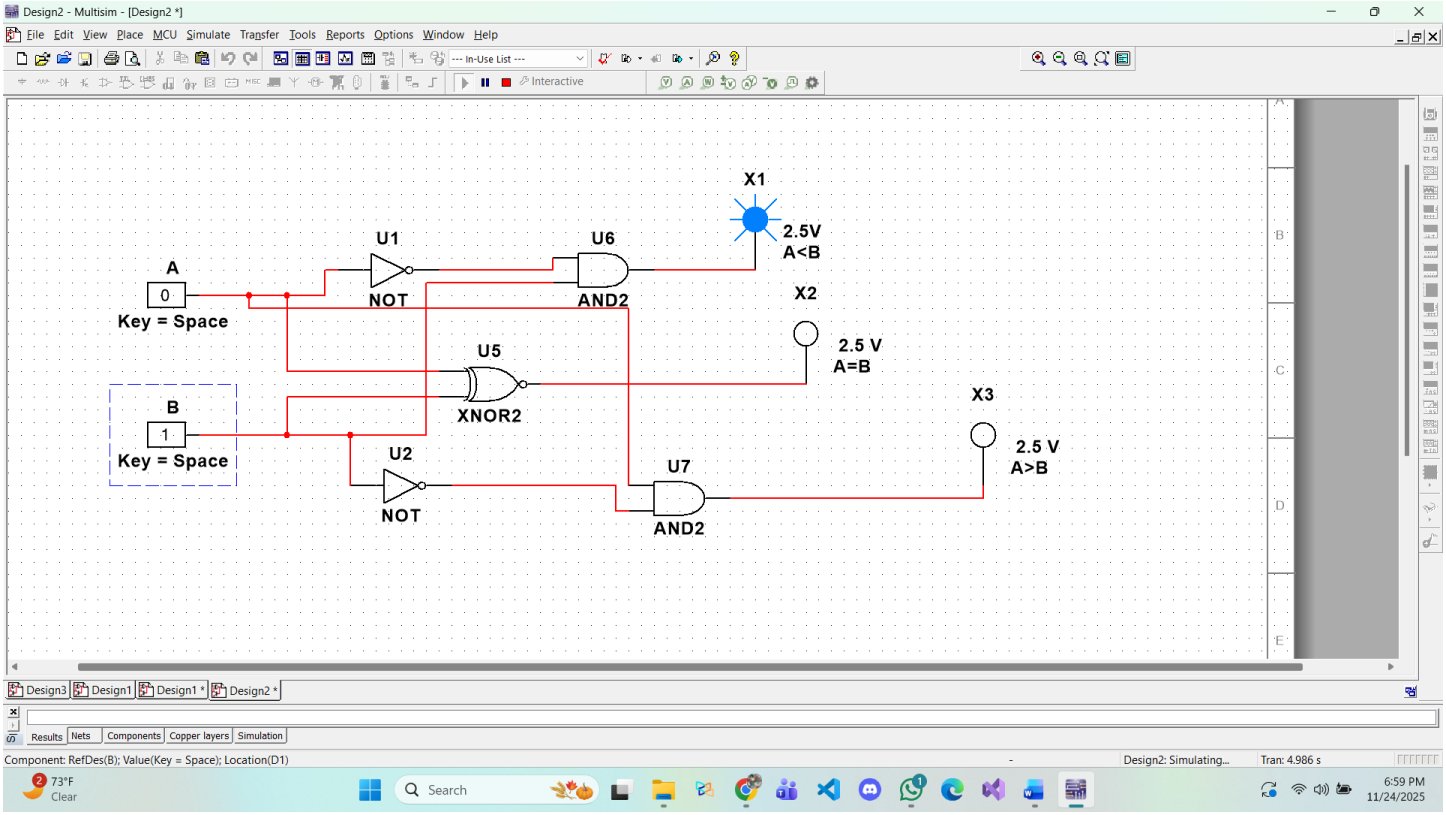


Fig: 1bit Comparator (A<B)

Discussion:

In this experiment, essential combinational circuits including the half adder, full adder, and 1-bit comparator were designed and tested. The half adder was implemented to perform binary addition of two single-bit inputs, producing sum and carry outputs using XOR and AND gates. The full adder extended this by handling an additional carry-in input, allowing multi-bit addition functionality. Truth tables were constructed for both adders and verified through simulation and hardware implementation. The 1-bit comparator was designed to compare two single-bit binary inputs and determine if one is greater than, less than, or equal to the other. Each circuit was simulated in NI Multisim and confirmed through practical testing.

Conclusion:

The experiment effectively demonstrated the construction and analysis of basic arithmetic and comparison circuits. The successful implementation helped reinforce key digital logic design concepts.

Reference:

- CircuitsToday, "Half Adder and Full Adder," [Online]. Available: http://www.circuitstoday.com/half-adder-and-full-adder. [Accessed: Nov. 24, 2025].
- Wikipedia, "Comparator," [Online]. Available: https://en.wikipedia.org/wiki/Comparator#:~:text=In%20electronics%2C%20a%20comparator%20is,and%20one%20binary%20digital%20output%20. [Accessed: Nov. 24, 2025].